Albin Andrews Joby

Electrical and Electronic Engineering student

Stage 2 BE Electrical and Electronic Engineering student at University College Dublin. Co-Founder and Head of Electrical and Electronics at GaelForce UCD, Ireland's first VEX U robotics team, and an Electrical Engineer at UCD Formula Student working on the accumulator. Focus: embedded systems, PCB design, motor control, 3D printing and rapid prototyping. I'm extremely curious about new technologies, and I am quick to take every chance to learn new methods, tools and software. I also have a strong interest in new developments in AI.

EDUCATION

BE Electrical and Electronic Engineering
09/2025 - Present

University College Dublin, Belfield

Stage 2 · GPA 3.47

Leaving Certificate · 589 points
2025

St Aidan's CBS, Dublin 9 · Vice-Chairman of the Green Schools Committee; helped earn the school's first Green Flag for Global Citizenship & Energy

PROJECTS

Co-Founder and Head of Electrical and Electronics
GaelForce UCD

04/2026 - Present

See the build on my portfolio →

- Co-founded Ireland's first VEX U robotics team and grew it from 6 founding members to 23 members and 12 sponsors; lead the Electrical and Electronics department and the robot's autonomous systems.
- Designed and built a fully custom ESP32-S3 odometry and localisation pod that fuses two AMT102-V encoders, a BNO085 IMU and four TMF8821 time-of-flight sensors, sending the robot's pose to the VEX V5 brain over RS-485 at 100 Hz.
- Took the pod from a working breadboard prototype through two KiCad PCB revisions, fabricated in the UCD Elecworkshop and hand-soldered; redesigned the V2 pin map around cable routing.
- Bench-tested the V2 board, found and hand-fixed three layout faults, and brought up a working link to the V5 brain.
- Personally secured three 3D-printing sponsors (SUNLU, 3DJake and eSUN) and helped win the backing of the UCD School of Electrical and Electronic Engineering.

Next: testing the localisation firmware on the V2 board. The team is aiming for the VEX World Championship in Las Vegas.

CONTACT

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github.com/albinjoby82-ops

PORTFOLIO

myportfolio.albinjoby82.workers.dev →

Photos, CAD, PCBs and write-ups of every project

SKILLS

Always ready to take a project or team from start to finish.

Embedded Systems
Embedded C, C/C++, ESP32 (S3, C6), ESP-IDF / PlatformIO, Arduino, State Machines, Sensor Data Logging, MATLAB

Electronics and PCB Design
KiCad, Schematic and PCB Design, Component Selection, Analogue and Digital Electronics, LTspice, Logisim Evolution, RS-485, I^2C, UART, Sensor Integration (IMU, ToF, Encoders)

Lab and Manufacturing
PCB Manufacturing, Soldering, PCB Assembly and Rework, Bench Testing and Debugging, Oscilloscope, Signal Generator, Multimeter, Bench Power Supply

CAD and Rapid Prototyping
Fusion 360, Onshape, 3D Printing, Bambu Studio, Laser Cutting, xTool

PROJECTS (CONTINUED)

Electrical Engineer, High-Voltage Systems

UCD Formula Student (UCDFS) — 09/2026 - Present

- Part of the team designing a new accumulator (the car's high-voltage battery pack) from scratch this season; design in progress.
- Working on the accumulator's cell placement and manufacture, with every accumulator part to be 3D printed in flame-retardant (FR) filament.
- Designed custom busbars, sourced them through supplier outreach, and sourced a spot welder for welding the cells to the busbars.
- Researched busbar and high-voltage connection design against the Formula Student rules to guide a compliant design.
- Currently repairing last season's accumulator so the car can run for two master's theses in late November 2026.

Event Director

Dublin Micromouse Open, hosted by UCD ElecSoc — 09/2026

See the event on my portfolio →

- Led the organising team and set every target and direction for a one-day competition where 20 teams and 110 participants built and raced autonomous maze-solving robots.
- Designed the format so no prior experience was needed: an identical kit for every team, a six-hour build, then timed races on a 16 × 16 maze.
- Led the design of the mouse around an ESP32-C6, three VL53L0X distance sensors, an IMU and two N20 encoder motors, and built and presoldered 25 kits with the team.
- Wrote a debug kit of single-component pass/fail tests ending in a flood-fill maze solver, plus the step-by-step help section of the event website.
- Designed the sponsorship deck that brought in Anthropic as prize sponsor, with Red Bull, the UCD School of Electrical and Electronic Engineering and the UCD Innovation Academy as in-kind backers.

WORK EXPERIENCE

Assistant Manager

Gala, Collins Avenue, Dublin — 2022 - 2026

- Ran shifts and supervised staff, including cashing up, inventory, stock control and purchasing.
- Handled customer service and daily store operations; rehired across three separate periods.

MORE SKILLS

Software
Git, GitHub, TypeScript, React, Web Development, Linux, Cloudflare Workers, Firebase, AI-Assisted Development (Claude Code / Codex)

Leadership
Team Leadership, Sponsorship and Fundraising, Event Direction, Recruitment, Technical Writing, Workshop Delivery

SEE THE WORK

Portfolio
myportfolio.albinjoby82.workers.dev

GaelForce and the odometry pod
/projects/gaelforce-ucd

Odometry pod write-up, KiCad files and Gerbers
github.com/albinjoby82-ops

Dublin Micromouse Open
/projects/dublin-micromouse-open